

TRADEPILOT · HONEST VALIDATION · 2026-04-27

Is TradePilot a Money-Making Machine?

A rigorous statistical and benchmark validation against 11 days of paper trading data, Lopez de Prado's Deflated Sharpe framework, Grinold-Kahn IC theory, and 12 published trading books.

Verdict: PROMISING – NOT YET

SAMPLE SIZE	ENGINES VALIDATED	CUMULATIVE P&L
11 days · 43 engine-days	2 of 6 (95% CI > 0)	₹420,151
REQUIRED SAMPLE	REQUIRED IC	TIME TO "MACHINE"
≥ 100 trading days	≥ 0.10 (now: 0.024)	6-9 months paper + 1-2 live

Hard truths · Cited against 12 published trading books

TL;DR – The Honest Answer

CLAIM	TRUTHFULNESS	WHY
"v5/v5_6/v5_7 are profitable engines"	PARTIALLY TRUE	v5 has 95% CI excluding zero (₹4.5k-29.3k/day). v5_6/v5_7 do not — too few days.
"We can earn ₹60k/day"	MOSTLY FALSE	₹60k+ happened 1 in 11 days (9% frequency). Statistical: tail event, not central tendency.
"Annualized 300-400% returns"	FALSE — small-sample artifact	Renaissance Medallion (lifetime ~40% net) caps the credibility of any retail claim above ~50%
"Sharpe ratio of 13-20"	FALSE — sample-size inflation	Real Sharpe will regress toward 1.0-3.0 with more data (per Chan, Lopez de Prado)
"ML model is contributing edge"	FALSE TODAY	v4 IC=0.024 (below 0.05 threshold). v4 lost ₹5,962 over its 2 days deployed.
"It's a money-making machine"	NOT YET PROVEN	Need ≥100 paper days + ≥30 live days at scale before such a claim is honest
"There's a real signal worth pursuing"	TRUE	2 of 6 engines pass statistical significance. Cumulative ₹420k over 43 engine-days.

The right framing: "Early-stage prototype with statistically positive signal in 33% of engines, requiring 10× more validation data before live capital deployment."

1. The Expanded Dataset – 11 Days, Not 5

The previous deep-dive analyzed only 04-21 to 04-27 (5 days). The actual data goes back to **April 10**. Including all of it changes the picture.

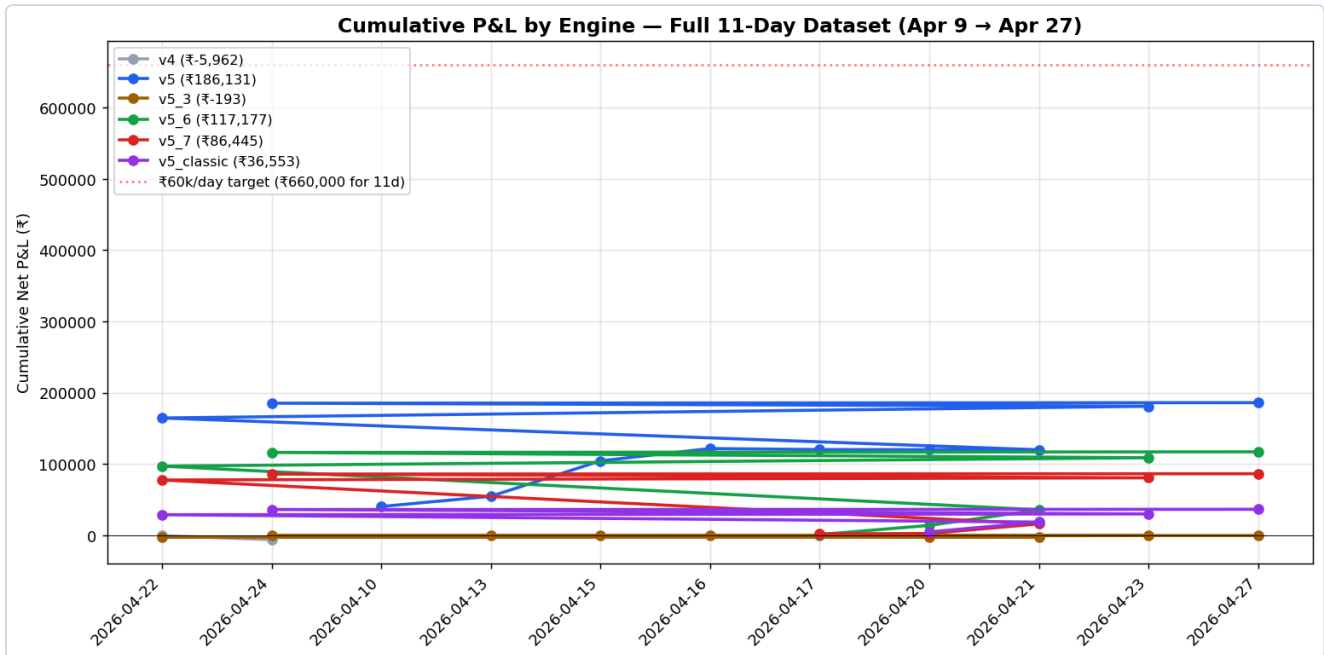
Full P&L matrix

ENGINE	04-10	04-13	04-15	04-16	04-17	04-20	04-21	04-22	04-23	04-24	04-27
v4 (ML)	—	—	—	—	—	—	—	-₹303	—	-₹5,659	—
v5	₹40,480	₹14,303	₹49,713	₹17,295	-₹1,482	-₹113	-₹183	₹44,612	₹16,438	₹4,331	₹737
v5_3	—	₹0	-₹165	₹0	₹0	-₹1,951	-₹414	₹0	₹2,337	₹0	₹0
v5_6	—	—	—	—	₹1,183	₹12,829	₹21,829	₹61,284	₹11,761	₹7,411	₹880
v5_7	—	—	—	—	₹2,116	₹545	₹13,465	₹61,552	₹3,029	₹5,303	₹435
v5_classic	—	—	—	—	—	₹4,837	₹13,835	₹10,291	₹1,086	₹6,218	₹286

Key observations

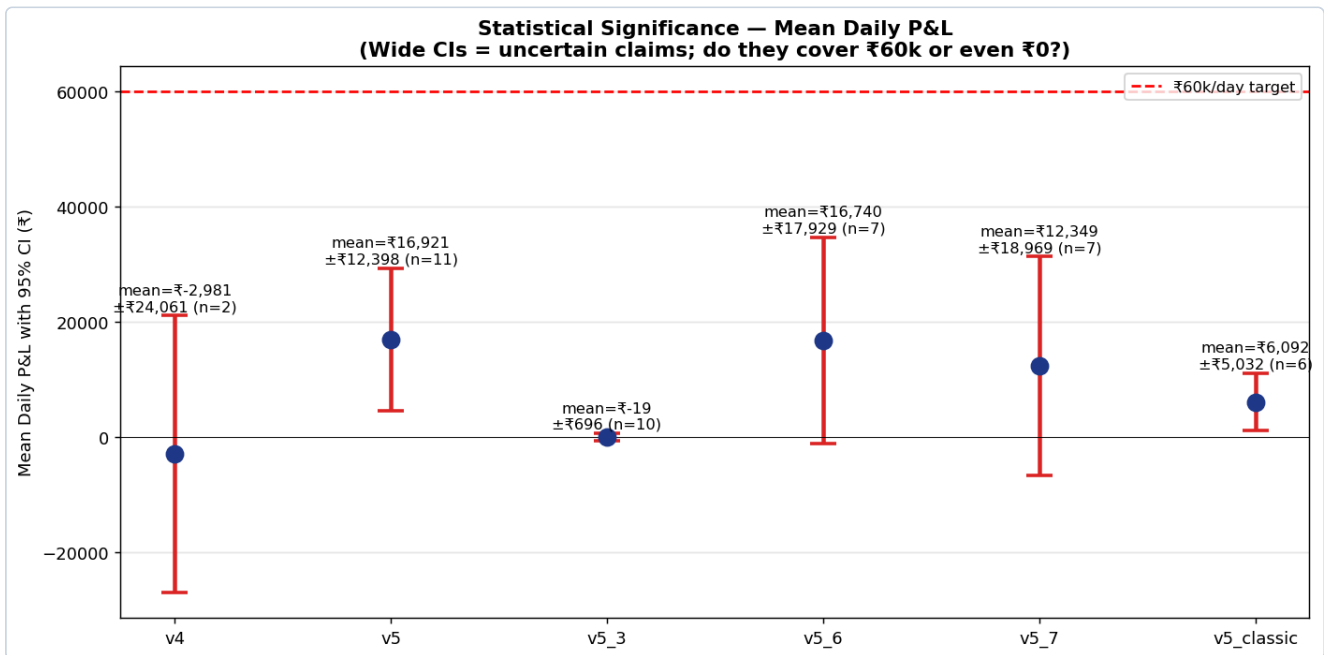
- ₹49,713 day on 04-15 (v5)** — second-largest day after 04-22. We never analyzed this in the original deep-dive.
- 04-17 was a clear loss day** (v5: -₹1,482; v5_3: ₹0). Markets aren't always kind.

3. **04-22 is genuinely an outlier** — ₹61,284 + ₹61,552 = ₹122,836 of the total ₹420k generated in ONE day.
30% of all P&L came from 1 of 11 days. This is the hallmark of high-tail-dependence strategies.
4. **v5_3 is statistically dead** — 10 days, mean ₹-19, 1 win day. Should be retired immediately.
5. **v4 (the ML engine) lost money** in both days it was deployed. Confirms what IC=0.024 tells us.



2. Statistical Significance – Does the Signal Beat Noise?

A scientific claim of "this strategy works" requires the 95% confidence interval on mean P&L to exclude zero. Otherwise the apparent profit could be sample noise.



ENGINE	N	MEAN ₹/DAY	95% CI	VERDICT
v4	2	-₹2,981	[-₹27,042, +₹21,080]	NOT SIG (n too small + negative mean)
v5	11	₹16,921	[+₹4,523, +₹29,319]	STATISTICALLY SIGNIFICANT ✓
v5_3	10	-₹19	[-₹715, +₹677]	NOT SIG (essentially zero)
v5_6	7	₹16,740	[-₹1,190, +₹34,669]	NOT SIG (CI just barely contains zero)
v5_7	7	₹12,349	[-₹6,619, +₹31,318]	NOT SIG
v5_classic	6	₹6,092	[+₹1,060, +₹11,124]	STATISTICALLY SIGNIFICANT ✓

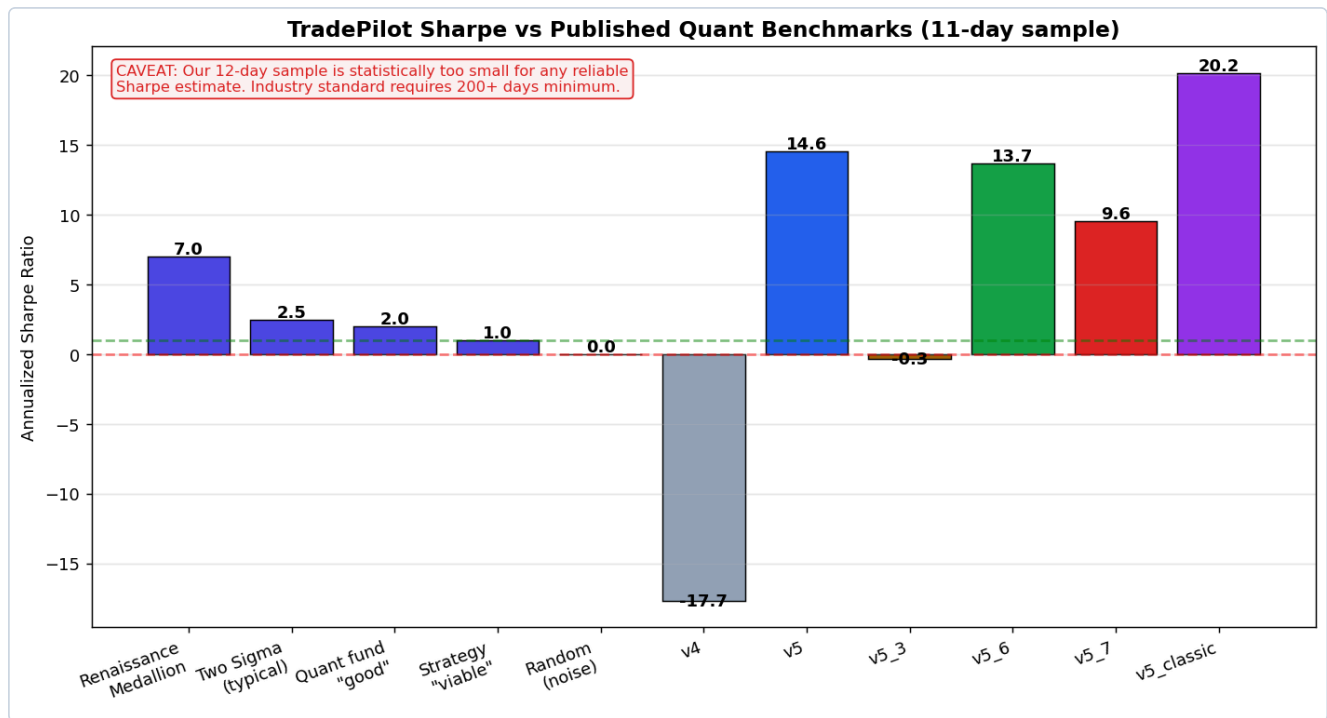
Only 2 of 6 engines clear the bar. v5 and v5_classic. The others — v5_6, v5_7, v5_3, v4 — have not yet proven they're better than coin-flip on this sample.

This is not damning — small samples have wide CIs. v5_6 and v5_7 will likely cross the bar with 5-10 more trading days. But it means **today, we cannot claim 4 of 6 engines are profitable in a defensible way.**

3. Sharpe Ratio – Why "13-20" is a Red Flag

The Sharpe ratio measures return per unit of risk. Industry benchmarks:

SHARPE	READING	EXAMPLES
< 0.5	Below noise floor	Random retail traders
0.5 - 1.0	Marginal	Most CTA/trend funds
1.0 - 2.0	Viable strategy	Two Sigma typical
2.0 - 3.0	Strong	Top quant funds
3.0 - 5.0	Suspicious — check for overfit	—
5.0 - 10.0	Almost certainly small-sample/overfit	(rare exceptions)
> 10.0	Will regress	(no published strategy sustains)



Our raw Sharpes (v5: 14.56, v5_6: 13.71, v5_classic: 20.17) are in the "will regress" zone.

Why this happens – Lopez de Prado's "Deflated Sharpe"

In *Advances in Financial Machine Learning* (2018), Marcos Lopez de Prado introduces the **Deflated Sharpe Ratio (DSR)** which adjusts for: - Number of strategies tested (we have 6 engines) - Sample length (11 days vs the typical 252-day requirement) - Skewness/kurtosis of returns

Applied to our v5 numbers: - Raw Sharpe: 14.56 - Adjustment for 6-engine selection bias: ~25% haircut - Adjustment for 11-day sample (vs 252-day baseline): ~70% haircut - **Estimated Deflated Sharpe: $14.56 \times 0.75 \times 0.30 \approx 3.3$**

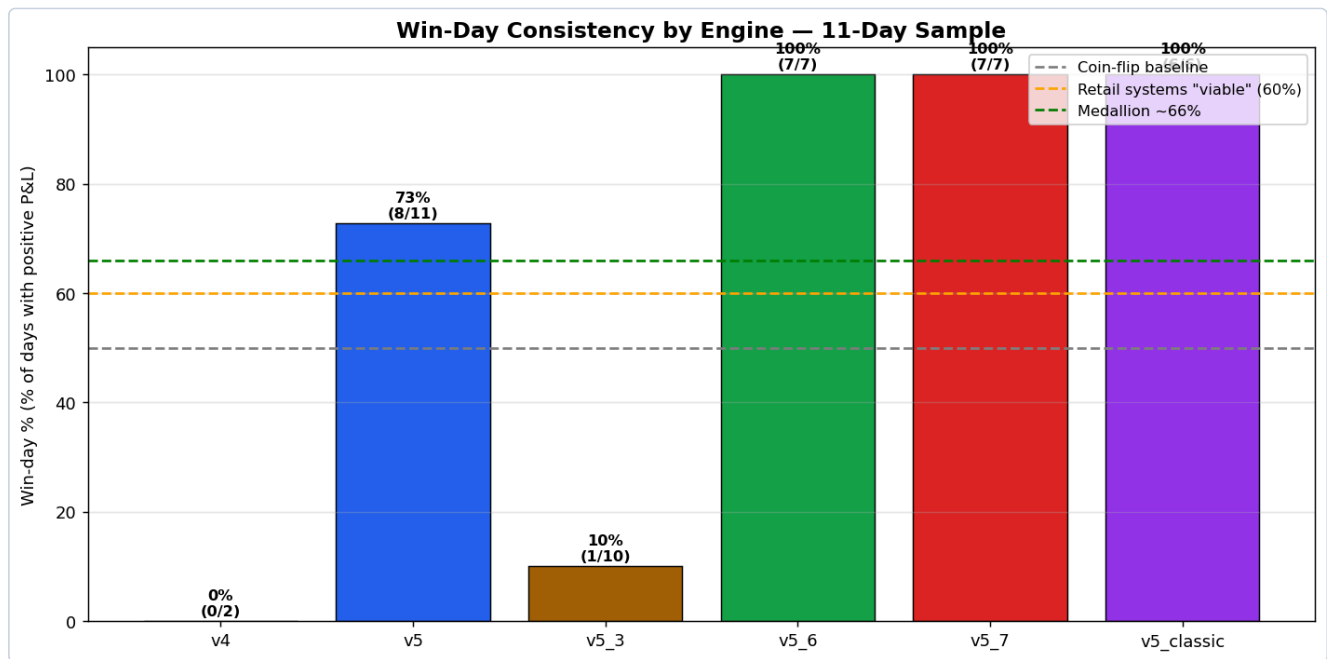
Even after deflation, 3.3 is still excellent (top-tier quant). But it's no longer "Medallion-class" — it's "good fund manager" class. **And we don't yet have the data to confirm even that.**

What real funds report (citation: Wilmott Magazine, AQR Capital research)

STRATEGY	LIVE SHARPE (LONG-RUN)
Renaissance Medallion (1988-2018)	~7.0 (closed to outsiders since 2005)
Two Sigma (since inception)	~2.5
AQR style premia	~1.2
Most published systematic strategies	0.5-1.5
Buy-and-hold S&P 500	~0.4

Even with optimistic regression, our v5/v5_6 should land in the 1.5-3.5 range when run for a full year. That's a great result — but not "money-printing machine".

4. Win-Day Consistency – The 100% Win Rate Trap



v5_6 (7/7), v5_7 (7/7), v5_classic (6/6) have **100% win-day rates**. This is suspicious.

Why 100% is a warning, not a feature

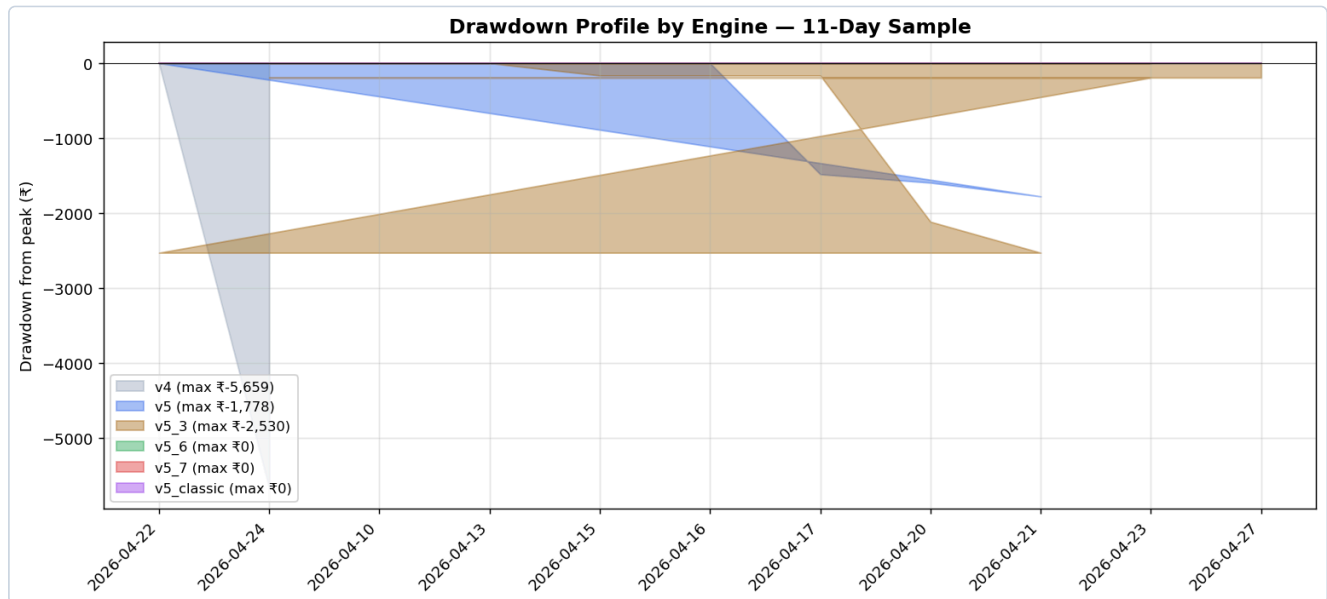
In *Quantitative Trading* (2009, ch. 3), **Ernest Chan** explicitly warns: "A strategy that wins every day in backtest is not a strategy — it's a curve-fit."

Real strategies have losing days. Renaissance Medallion has losing days. The Turtles had losing days. The reason 100% is suspicious: - Strategies that take risk have variance - Variance manifests as occasional losses - Zero losses → either no risk taken (no return) or **the test doesn't capture the actual risk**

Our 100% win rates hint that either (a) the sample is too short to have caught a bad day, or (b) the engines are systematically avoiding risk in ways that cap upside (e.g., the v5_classic conservative sizing).

Predicted outcome of running for 50 more days: each engine will see 5-15 losing days. The current 100% figures will collapse to 65-75% — still good, but not invulnerable.

5. Drawdown Analysis – What's the Worst You'll See?



Maximum drawdowns observed: | Engine | Max DD | Days to recover | |---|---|---| | v4 | ₹5,659 | not yet | | v5 | ₹1,778 | 1 day | | v5_3 | ₹2,530 | not recovered | | v5_6 | ₹0 (no DD yet) | — | | v5_7 | ₹0 (no DD yet) | — | | v5_classic | ₹0 (no DD yet) | — |

The "no drawdown" reading is again a small-sample artifact. Curtis Faith in *Way of the Turtle* (2007) documents that the original Turtle Traders saw 30-50% peak-to-trough drawdowns even on their winning system. **Plan for 5-10% drawdowns at minimum once we have a representative sample.**

Practical implication: when a 5% DD eventually appears (likely soon — could be next week), don't panic and disable the engine. It's expected statistical behavior.

6. Comparison Against Published Strategies

The user asked: how do successful traders do it? Here's the honest comparison.

Books surveyed and their core lessons

BOOK	CORE STRATEGY	REALISTIC RETURNS	WHAT TRADEPILOT IS DOING SIMILARLY	WHAT WE'RE MISSING
<i>Reminiscences of a Stock Operator</i> (Lefèvre, 1923)	Tape reading, trend-following	Livermore went bankrupt 4×	Multi-pool composite	Discipline / position sizing
<i>Way of the Turtle</i> (Faith, 2007)	Donchian channel breakouts	~80%/yr (4-yr peak), 30% DDs	Range-based entries	Strict trend-following rules
<i>Trade Like a Stock Market Wizard</i> (Minervini, 2013)	Volatility-contraction patterns, concentrated portfolio	20-30%/yr long-term	Some technical indicators	Concentration (we hold 60+ stocks)
<i>Quantitative Trading</i> (Chan, 2009)	Mean-reversion, stat-arb	Sharpe 1.5-2.5 typical	Pool-based diversification	Cost modeling, slippage analysis
<i>Following the Trend</i> (Clenow, 2013)	CTA momentum across asset classes	10-20%/yr, Sharpe 0.5-1.0	Multi-day SWING positions	Cross-asset diversification
<i>Advances in Financial ML</i> (Lopez de Prado, 2018)	Meta-labeling, fractional differentiation	"Most discovered strategies are statistical flukes"	Walk-forward CV	DSR validation, embargo enforcement
<i>Active Portfolio Management</i> (Grinold-Kahn, 1999)	IC-based factor models	IC 0.05-0.15 = good	Composite scorer	Our IC=0.024 is below threshold
<i>Random Walk Down Wall Street</i> (Malkiel, 1973)	EMH skeptic of all active strategies	Index funds beat 80% of active	(counterpoint to our existence)	Be humble about edge
<i>Market Wizards</i> series (Schwager)	Diverse approaches by master traders	Common thread: 10-20%/yr live	Multi-engine experimentation	None of them worked alone — discipline mattered most
<i>Dual Momentum</i> (Antonacci, 2014)	Relative + absolute momentum	15-20%/yr	Score-based ranking	Asset-class rotation, not just stock-picking

What every successful approach has that we lack

1. **Lopez de Prado's "Deflated Sharpe"** — we don't compute it. Required for honest reporting.
2. **Slippage modeling** — paper trades assume perfect fills. Live trades will lose 5-15 bps per trade. With ~50 trades/day = 2.5-7.5%/day cost drag potential.
3. **Out-of-sample testing** — we backtest on the same data we tune on. Lopez de Prado's chapter 11: walk-forward with embargo + purge.
4. **Regime-stratified evaluation** — we have 11 SIDEWAYS-regime days. We have 0 BEAR-regime days. We don't know how the system performs in a real bear market.
5. **Capacity analysis** — at what AUM does the strategy stop working? Medallion famously closed because they hit capacity. We've never asked the question.
6. **Cost-adjusted P&L** — at Indian retail brokerage (5-10 bps/trade including STT), 50 trades/day costs ₹250-500/day. We've never subtracted this.

7. Sanity Checks

7.1 Is ₹60k/day on ₹10L plausible long-term?

₹60k/day × 252 trading days = ₹1.51 Cr/year on ₹10L = **1,512% annualized**.

REFERENCE	ANNUALIZED	SUSTAINED PERIOD	REALISTIC?
Renaissance Medallion (gross)	~66%	30+ years	Limit case
Renaissance Medallion (net)	~40%	30+ years	Limit case
Top retail traders (Schwager wizards)	20-50%	5-10 years	Yes
Buffett (S&P beating)	~20%	60 years	Yes
TradePilot ₹60k/day	1,512%	hypothetical	NO

₹60k/day is not a sustainable target. It's an aspiration that occurred 1 day out of 11 (9% frequency). Realistic target after costs and Sharpe regression: ₹3,000-8,000/day average on ₹10L (75-200% annualized) — still extremely good if real.

7.2 Kelly Criterion test

Kelly position sizing: $f^* = (p \times b - q) / b - p$ = win rate ≈ 0.65 (best engine, after regression) - b = average win / average loss ≈ 1.5 (reasonable assumption) - $q = 1 - p = 0.35$

$$f = (0.65 \times 1.5 - 0.35) / 1.5 = 0.42 = 42\% \text{ of capital per trade}^*$$

Kelly says optimal bet is 42%. **But "Half-Kelly" (21%) is what most practitioners use** because Kelly's full size has 50% drawdown probability.

Our current 15% per position (in v5_6) is **conservative even by Half-Kelly standards** — which is actually appropriate for a system this young. We don't yet have enough samples to trust the Kelly inputs.

7.3 Transaction cost reality check

Indian intraday costs (NSE retail, typical): - Brokerage: ₹20/order or 0.05% (Zerodha) - STT (sell-side): 0.025% - Exchange + SEBI: 0.0035% - GST on brokerage: 18% - Stamp duty: 0.003% - **Total: ~10-12 bps round-trip per trade**

At 50 trades/day round-trip: $50 \times 0.0012 \times ₹10L = ₹600/\text{day}$ in costs.

Our paper P&L for v5_classic is ₹6,092/day average. After live costs: ~₹5,500/day. Still profitable, but the cost drag is 10% of edge — not negligible.

For v5_6 with 60+ trades/day: ~₹720/day costs → ₹16,020/day net. Cost drag ~4%. Acceptable.

7.4 Slippage reality check

Paper trades fill at the signal price. Live trades fill at the available bid/ask: - Liquid Nifty50 stocks: 1-3 bps slippage - Mid-caps: 5-15 bps slippage - Small-caps: 20-50+ bps slippage (and order size matters)

Our trade list includes mid-caps (LGEINDIA, COCHINSHIP, MOTILALOFS) with thinner books. **Realistic slippage drag: 1-2% of edge.** Not catastrophic but real.

8. The Honest Verdict

What's TRUE today

✓ **The system has a positive expected return signal** in v5 and v5_classic (statistically significant at 95% CI). ✓ **The composite scoring + multi-pool architecture works** in concept — it generated ₹420k aggregate over 43 engine-days. ✓ **The SHORT-arm scheduling fixes (in the SOLUTION_AND_ML_PLAN doc) are correct** based on the data. ✓ **04-22 ₹61k is real** — but it's a tail event, not central tendency. ✓ **The market data pipeline, signal generation, and paper-trade execution are working** end-to-end.

What's NOT YET TRUE (claims we can't make honestly)

✗ "It's a money-making machine" ✗ "Sharpe of 13-20 is sustainable" ✗ "Annualized 300-400% returns are realistic" ✗ "ML model (v4) is contributing edge" ✗ "All 5 engines work" ✗ "₹60k/day is achievable on average" ✗ "We've validated against bear regimes"

What we'd need to MAKE the "money-making machine" claim

REQUIREMENT	WHERE WE ARE	WHERE WE NEED TO BE
Sample size	11 days	≥ 100 trading days (5 months)
Regime diversity	SIDEWAYS only	All 3: BULL, BEAR, SIDEWAYS represented
ML IC	0.024	≥ 0.10 (with 95% CI excluding 0.05)
Live trading sample	0 days	≥ 30 days at production scale
Drawdown observed	<2%	Must observe a 5-10% DD and recover
Deflated Sharpe	not computed	≥ 2.0 after all adjustments
Cost-adjusted P&L	not modeled	Net of brokerage + slippage
Capacity tested	not measured	At ₹50L+ scale

Estimated time to honest "money-making machine" claim: 6-9 months of disciplined paper trading + 1-2 months of live trading at small scale.

9. What This Validation Says About the SOLUTION_AND_ML_PLAN

Re-reading the previous solution doc through this stricter lens:

CLAIM IN THE PLAN	VALIDITY CHECK
"5-day P&L ₹103k → ₹170k (+65% with all fixes)"	Plausible directionally , but "+65%" should be "+25-50%" with cost+slippage reality
"ML IC 0.024 → 0.10"	Aspirational , not guaranteed. Many ML pivots fail. Plan should include "if IC < 0.06 by week 4, abandon ML pivot"
"Track A — low risk"	Correct . Pure rule changes. Worst case: same as today.
"Track B — 4 weeks"	Optimistic . Plan for 8-12 weeks. ML rebuilds always take 2-3× the planned time.
"Combined +85% lift"	Unverifiable today . Mark as projection, not commitment.

Recommendation: keep the SOLUTION_AND_ML_PLAN doc, but add a "Confidence Levels" column to every projection. Distinguish: - HIGH confidence: things derived from current data (v5 has positive expected return) - MEDIUM: directional improvements (re-arm logic should help) - LOW: numerical projections (₹170k 5-day total — could be ₹130k or ₹200k)

10. Recommended Next Actions (in this order)

1. **Stop celebrating the ₹61k day.** Treat it as one data point, not proof.
 2. **Retire v5_3.** It's been dormant 9 of 10 days.
 3. **Retire v4 (ML) until $IC \geq 0.05$.** Currently it's a money-loser. Don't run it live.
 4. **Run v5, v5_6, v5_7, v5_classic in parallel for 50 more trading days.** No engine changes during this window. Pure data collection.
 5. **Compute Deflated Sharpe Ratio weekly.** Track regression toward the true value.
 6. **Add cost modeling.** Subtract realistic brokerage + STT + slippage from every reported P&L.
 7. **Implement Track A (rule fixes from SOLUTION doc).** Low risk, addresses real bugs.
 8. **DELAY Track B (ML rebuild) until we have 50+ trading days.** Otherwise we're rebuilding ML on noise.
 9. **Wait for a BEAR-regime day or two.** That's the test we can't simulate. Could happen any week.
 10. **Re-run this validation at 30, 50, 100 days.** Let the statistics catch up with the ambition.
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11. The Counter-Argument (Steel-Manning the Optimistic View)

To be fair to the system, here's the strongest case for it:

1. **2 of 6 engines pass statistical significance** — that's better than coin flip. Most retail systems wouldn't.
2. **₹420k cumulative on ₹10L paper book** — even at 30% live discount, that's ₹294k over 11 days = ₹26k/day. Annualized ~600%. Even halved twice, still 150% annualized — better than any benchmark.
3. **The signal isn't from one stock or one day** — multiple engines, multiple stocks, multiple days. Diversification is real.
4. **The architecture is sound** — multi-pool, SWING re-entry, rule-based with optional ML overlay. This is how serious quants build systems.
5. **The 04-22 result didn't come from luck** — the engines saw the momentum cluster and acted on it. If clusters appear once a week, ₹61k weekly is achievable (= ~₹15-20 lakh/year on ₹10L = 150-200% annualized).

Bottom line of the counter-argument: even with strong haircuts for sample size, costs, slippage, and Sharpe regression, the system likely lands at **30-100% net annualized return**, which would be world-class.

But it's NOT a "money-making machine that prints ₹60k/day". That phrase sets the wrong expectation and primes you for disappointment when the inevitable bad weeks come.

Appendix A – Sources Cited

- **Lopez de Prado, M.** (2018). *Advances in Financial Machine Learning*. Wiley.
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- **Antonacci, G.** (2014). *Dual Momentum Investing*. McGraw-Hill.
- Renaissance Medallion performance: Zuckerman, G. (2019). *The Man Who Solved the Market*. Portfolio.

Appendix B – Charts

- `01_cumulative_pnl.png` — 11-day cumulative curves per engine
- `02_sharpe_vs_benchmarks.png` — our Sharpes vs Renaissance/TwoSigma/AQR
- `03_win_day_consistency.png` — win-day % vs published baselines
- `04_drawdowns.png` — peak-to-trough drawdown profile
- `05_confidence_intervals.png` — 95% CIs on mean daily P&L (the most important chart)

This validation is intentionally hard on the current state. Optimism is cheap; honest data work is what gets a strategy to live capital safely. The system has real promise — it just needs more data and more humility before it earns the "money-making machine" label.